

Ex: 2

Date:

## PaaS in Public Cloud Using Render

### Objective

The purpose of this exercise is to build and deploy a web application on Render, a Platform-as-a-Service (PaaS) solution, to gain insights into how cloud providers host applications without the need for manual server administration.

### Understanding PaaS

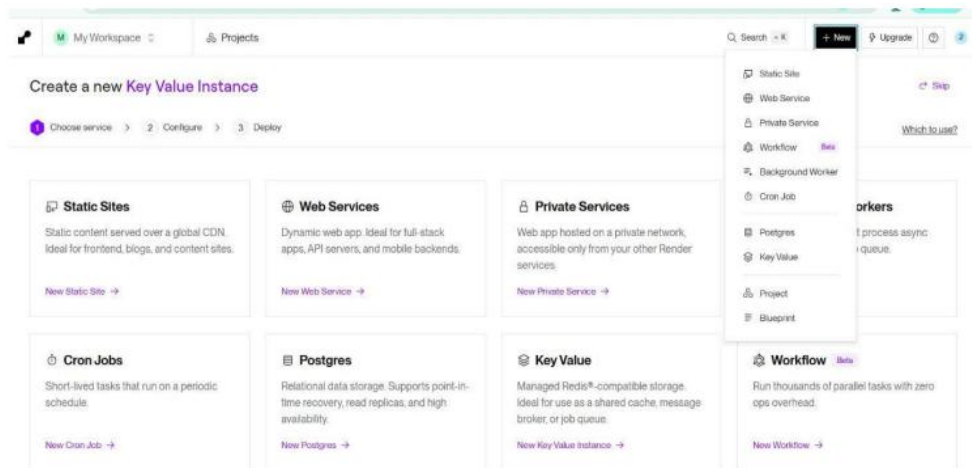
Platform-as-a-Service (PaaS) refers to a cloud computing model that offers a pre-configured environment for developers to create, execute, and oversee applications seamlessly.

### Choosing Render

Render serves as a specialized cloud hosting platform that streamlines the process of launching web applications.

### Methodology

1. Ensure you have an active GitHub account.
2. Establish a local project directory titled "RenderHelloApp."
3. Within this directory, generate two files: app.py and requirements.txt.
4. Input the Flask-based application code into the app.py file.
5. Populate requirements.txt with the necessary packages (Flask and gunicorn).
6. Create a public GitHub repository named "RenderHelloApp."
7. Upload your project files to the repository and commit the changes.
8. Register for a Render account and link it with your GitHub profile.



9. Initiate the creation of a new Web Service within the Render dashboard.
10. Select the

"RenderHelloApp" repository from your linked GitHub account.

## 11. Define deployment parameters:

- Language: Python
- Branch: main
- Root Directory: Leave blank
- Build Command: pip install -r requirements.txt
- Start Command: gunicorn app:app
- Instance Type: Free

The screenshot shows the 'New Web Service' form in the Render dashboard. The form is titled 'New Web Service' and includes a note: 'It looks like you're using Flask, so we've autofilled some fields accordingly.' The form fields are as follows:

- Source Code:** Thanaixemi / RenderHelloApp (5m ago) [Edit]
- Name:** RenderHelloApp (A unique name for your web service.)
- Language:** Python 3 (Choose the runtime environment for this service.)
- Branch:** main (The Git branch to build and deploy.)
- Region:** Singapore (Southeast Asia) (Your services in the same region can communicate over a private network.)
- Root Directory:** Optional (If set, Render runs commands from this directory instead of the repository root. Additionally, code changes outside of this directory do not trigger an auto-deploy. Most commonly used with a monorepo.)
- Build Command:** \$ pip install -r requirements.txt (Render runs this command to build your app before each deploy.)
- Start Command:** \$ gunicorn app:app (Render runs this command to start your app with each deploy.)
- Instance Type:** A table showing options for hobby and professional projects.

| For hobby projects                                                                                                                                                          | Free                   | 512 MB (RAM)<br>0.1 CPU | Upgrade to enable more features<br>Free instances spin down after periods of inactivity. They do not support SSH access, scaling, one-off jobs, or persistent disks. Select any paid instance type to enable these features. |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| For professional use<br>For more power and to get the most out of Render, we recommend using one of our paid instance types. All paid instances support:<br>• Zero Downtime | Starter<br>\$7 / month | 512 MB (RAM)<br>0.5 CPU |                                                                                                                                                                                                                              |
|                                                                                                                                                                             | Pro                    | 4 GB (RAM)              | Pro Plus<br>8 GB (RAM)                                                                                                                                                                                                       |

## 12. Execute the deployment and monitor the status until completion.

### Environment Variables

Set environment-specific config and secrets (such as API keys), then read those values from your code. [Learn more](#).

Generate

+ Add Environment Variable
+ Add from env

[Advanced](#)

Deploy Web Service

13. Navigate to the provided Render URL to verify the home page renders correctly.

Karthikeyan's work...

threatforge-backend v1

dep-da4m0uqtp3s73bt6td0

Search

+ New

Upgrade

Dashboard

threatforge-backend v1

Deploys

Settings

MONITOR

Events

Logs

Metrics

MANAGE

Environment

Shell

Compute

Previews

Disk

Changelog

Invite a friend

Contact support

Render Status

fix: route deployed frontend requests to backend

STATUS

Deploy succeeded | Live

DURATION

35.7s

DEPLOYED

Aug 22, 2026

TRIGGER

Auto-Deploy

SOURCE

g391667

All logs

Search logs

Aug 22, 3:57 PM - 3:20 PM

GMT+5:30

Aug 22

```

8", "content-length": 140}}, "responseTime": 5, "msg": "request completed"}

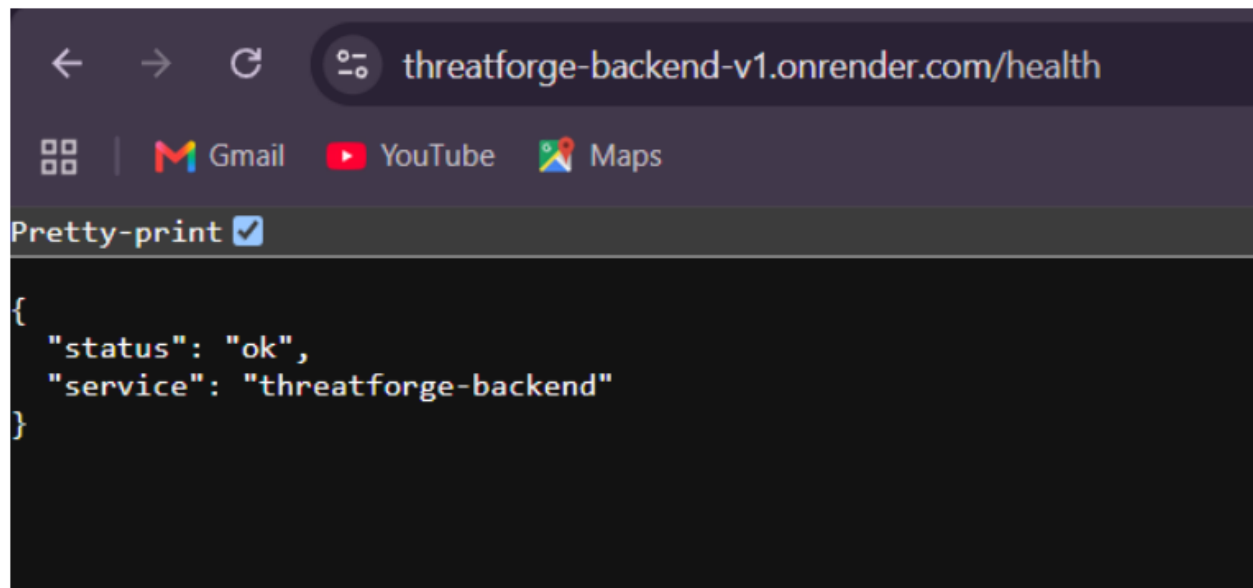
00:19:27 PM ==> Your service is live @!
00:19:27 PM [h3hg5] {"level":40,"time":1787392167911,"pid":86,"hostname":"srv-da4igu1500s73b9k2ag-hibernate-86f467cccb-h3hg5","req":{"id":2,"method":"GET","uri":"/","query":{"params":{"headers":{"host":"threatforge-backend-v1.onrender.com","user-agent":"Go-http-client/2.0","accept-encoding":"gzip, br","cdn-loop":"cloudflare; loops=1","cf-connecting-ip":"34.82.114.42","cf-ipcountry":"US","cf-ray":"a290f7b92b86755b-SEA","cf-visitor":{"scheme":"https"},"render-proxy-ttl":"4","rndr-id":"6399980a-1e90-4021","true-client-ip":"34.82.114.42","x-forwarded-for":"34.82.114.42, 172.68.23.199, 10.193.199.32","x-forwarded-proto":"https","x-request-start":"1787392167907212"},"remoteAddress":"::1","remotePort":40196},"res":{"statusCode":404,"headers":{"cross-origin-opener-policy":"same-origin","cross-origin-resource-policy":"same-origin","origin-agent-cluster":"","referrer-policy":"no-referrer","strict-transport-security":"max-age=31536000; includeSubDomains","x-content-type-options":"nosniff","x-dns-prefetch-control":"off","x-download-options":"noopen","x-frame-options":"SAMEORIGIN","x-permitted-cross-domain-policies":"none","x-xss-protection":"0","vary":"Origin","content-security-policy":"default-src 'none'","content-type":"text/html; charset=utf-8"},"content-length":139}}, "responseTime":1,"msg":"request completed"}

00:19:27 PM ==>
00:19:27 PM ==>
00:19:27 PM ==> Available at your primary URL https://threatforge-backend-v1.onrender.com
00:19:27 PM ==>
00:19:27 PM ==>

```

Need better ways to work with logs? Try the [Render CLI](#), [Render MCP Server](#), or set up a [log stream integration](#)

14. Validate the "/about" route by appending it to your application's URL.

A screenshot of a web browser window. The address bar shows the URL 'threatforge-backend-v1.onrender.com/health'. Below the address bar, there are icons for Gmail, YouTube, and Maps. A 'Pretty-print' button with a checkmark is visible. The main content area displays a JSON response: 

```
{
  "status": "ok",
  "service": "threatforge-backend"
}
```

```
← → ↻ threatforge-backend-v1.onrender.com/health
Gmail YouTube Maps
Pretty-print ✓
{
  "status": "ok",
  "service": "threatforge-backend"
}
```

## Conclusion

The exercise successfully demonstrated the development and deployment of a web application utilizing Render as a PaaS, effectively showcasing the ease of cloud-based application hosting.